

Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs

1. [ACJC/Prelims 2014/P1/1]

A student uses a digital ammeter to measure a current. The reading of the ammeter is found to fluctuate between 1.98 A and 2.02 A. The manufacturer of the ammeter states that any reading has a systematic uncertainty of $\pm 1\%$. Which value of the current should be quoted by the student?

- A (2.00 ± 0.01) A B (2.00 ± 0.02) A C (2.00 ± 0.03) A D (2.00 ± 0.04) A

2. [ACJC/Prelims 2014/P1/2]

An athlete of mass 80 kg competes in a 100 m race.

What is the best estimate of his mean kinetic energy during the race?

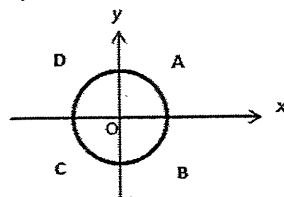
- A 4×10^2 J B 4×10^3 J C 4×10^4 J D 4×10^5 J

3. [ACJC/Prelims 2014/P1/3]

The figure below shows a circular path taken by a particle. At a certain instant, the particle's horizontal velocity, v_x , is $+2 \text{ m s}^{-1}$, and its vertical velocity, v_y , is -2 m s^{-1} .

Horizontally, right is taken to be positive x -direction.

Vertically, upward is taken to be positive y -direction.



Through which quadrant is the particle moving at this instant if it is travelling anti-clockwise around the circle?

4. [AJC/Prelims 2014/P1/1]

Which of the following pairs of physical quantities are both vector quantities?

- A work done, electric current
 B work done, electric field strength
 C force, electric current
 D force, electric field strength

5. [AJC/Prelims 2014/P1/2]

The diameter D of a sphere is measured to be 5.0 cm with a fractional uncertainty of 0.02.

What is the absolute uncertainty and fractional uncertainty of the radius R of the sphere?

	absolute uncertainty of R	fractional uncertainty of R
A	0.05 cm	0.01
B	0.1 cm	0.01
C	0.05 cm	0.02
D	0.1 cm	0.02

6. [AJC/Prelims 2014/P1/3]

The precision and accuracy of a measurement is affected by:

	precision	accuracy
A	systematic error	random error
B	random error	random error
C	systematic error	systematic error
D	random error	systematic error

7. [CJC/Prelims 2014/P1/1]

Which of the following quantities are paired with the correct units?

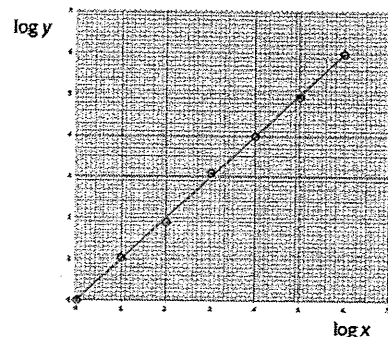
	quantity	units
A	work	newton (N)
B	e.m.f.	ampere (A)
C	current	coulomb (C)
D	pressure	pascal (Pa)

8. [CJC/Prelims 2014/P1/2]

Two physical quantities, x and y , are related by the equation,

$$y = kx^n$$

where n and k are non-zero positive integers. A series of measurements of x and y were carried out by a student, and the graph of $\log y$ against $\log x$ is plotted as shown below.



Which of the following statements best describes the data acquired by the student?

- A It is accurate and precise.
- B It is inaccurate but precise.
- C It is accurate but not precise.
- D It is inaccurate and not precise.

9. [DHS/Prelims 2014/P1/1]

Which quantity has different units from the other three?

- A rate of change of momentum
- B gradient of electric potential energy
- C impulse per unit time
- D power over displacement

10. [DHS/Prelims 2014/P1/2]

The Lyman series is the series of transitions resulting in ultraviolet emission lines of hydrogen atom as an electron goes from higher energy levels to ground state. It is given by the formula

$$\frac{1}{\lambda} = R_H \left(1 - \frac{1}{n^2} \right)$$

where n is a natural number greater than or equal to 2.

A student conducted an experiment to determine the constant R_H . The results are summarized below:

n	wavelength λ / nm
2	122 ± 2

What is the average value for R_H and its corresponding uncertainty?

- A $(1.093 \pm 0.009) \times 10^7 \text{ m}^{-1}$
- B $(1.09 \pm 0.02) \times 10^7 \text{ m}^{-1}$
- C $(1.09 \pm 0.04) \times 10^7 \text{ m}^{-1}$
- D $(1.1 \pm 0.1) \times 10^7 \text{ m}^{-1}$

11. [HCI/Prelims 2014/P1/1]

What is the best estimate for the value of the mass of a raindrop?

- A 4 μg
- B 4 mg
- C 4 g
- D 4 kg

12. [HCI/Prelims 2014/P1/2]

A student measures two lengths as follows:

$$X = 15.0 \pm 0.2 \text{ cm}$$

$$Y = 30.0 \pm 0.2 \text{ cm.}$$

The student calculates:

F_X the fractional uncertainty in X

F_Y the fractional uncertainty in Y

F_{Y-X} the fractional uncertainty in $(Y - X)$

F_{X+Y} the fractional uncertainty in $(X + Y)$

Which of these uncertainties has the largest magnitude?

- A F_X
- B F_Y
- C F_{Y-X}
- D F_{X+Y}

13. [JJC/Prelims 2014/P1/1]

The ideal gas law can be expressed in the equation

$$PV = nRT$$

where P is the absolute pressure of the gas, V is the volume of the gas, n is the amount of substance of gas (measured in moles), R is the molar gas constant, and T is the absolute temperature of the gas.

Which one of the following is the SI base unit for R ?

- A $\text{J K}^{-1} \text{mol}^{-1}$ B $\text{Pa m}^3 \text{K}^{-1} \text{mol}^{-1}$ C $\text{kg m}^2 \text{K}^{-1} \text{mol}^{-1} \text{s}^{-2}$ D $\text{kg}^2 \text{m}^3 \text{K mol}^{-1} \text{s}^{-2}$

14. [JJC/Prelims 2014/P1/2]

In a simple electrical circuit, the current in a resistor is measured as $(3.50 \pm 0.05) \text{ mA}$. The resistor is marked as having a value of $4.3 \Omega \pm 2\%$.

Using these values to calculate the power dissipated in the resistor, what would be the percentage uncertainty in the value obtained?

- A 3% B 4% C 5% D 6%

15. [JJC/Prelims 2014/P1/1]

Which estimate is realistic?

- A The temperature in the domestic refrigerator is 200 K.
B The surface area of a standard soccer ball is 0.15 m^2 .
C The current drawn by a laptop is 15 A.
D The volume of a pen is $4.0 \times 10^{-3} \text{ m}^3$.

16. [JJC/Prelims 2014/P1/2]

P and Q represent two forces.

Fig. 16.1, 16.2 and 16.3 represent three possible vector diagrams for P and Q .

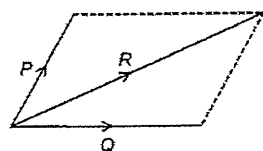


Fig. 16.1

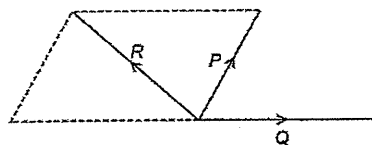


Fig. 16.2

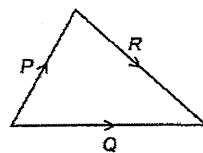


Fig. 16.3

Which of the following correctly represents the vectors R , S and T ?

	R	S	T
A	$P + Q$	$Q - P$	$P - Q$
B	$P + Q$	$P - Q$	$Q - P$
C	$P - Q$	$P + Q$	$Q - P$
D	$Q - P$	$P - Q$	$P + Q$

17. [MJC/Prelims 2014/P1/1]

In a simple electrical circuit, the current in a resistor is measured as $(2.50 \pm 0.05) \text{ mA}$. The resistor has a resistance of $4.7 \Omega \pm 2\%$.

If these values were used to calculate the power dissipated in the resistor, what would be the percentage uncertainty of the power?

- A 4% B 5% C 6% D 7%

18. [MJC/Prelims 2014/P1/2]

A metal sphere of radius r is dropped into a tank of water. As it sinks at speed v , it experiences a drag force F given by $F = kr^2v$, where k is a constant.

What are the SI base units of k ?

- A $\text{kg m}^2 \text{s}^{-1}$ B $\text{kg m}^{-2} \text{s}^{-2}$ C $\text{kg m}^{-1} \text{s}^{-1}$ D kg m s^{-2}

19. [NJC/Prelims 2014/P1/1]

The speed v of surface waves of wavelength λ on a liquid of density ρ is given by

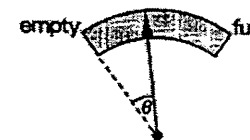
$$v = \left[\left(\frac{a\lambda}{2\pi} \right) + \left(\frac{2\pi b}{\rho\lambda} \right) \right]^{\frac{1}{2}}$$

where a is a constant and b is a quantity characteristic of the liquid. The dimensions for a and b in SI base units are, respectively,

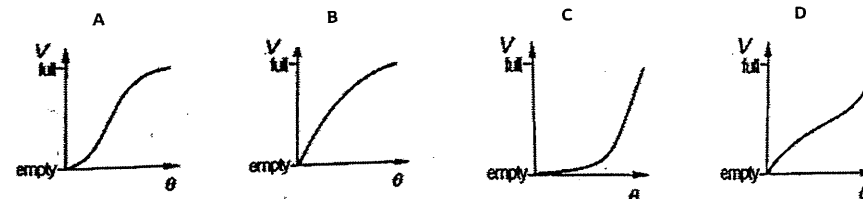
- A $\text{m}^{1/2} \text{s}^{-1}$ and $\text{kg}^{-1/2} \text{m}^2 \text{s}^{-1}$
B m s^{-2} and kg s^{-2}
C $\text{m}^2 \text{s}^{-2}$ and $\text{kg}^{-1/2} \text{m}^2 \text{s}^{-1}$
D m s^{-2} and kg m^{-2}

20. [NJC/Prelims 2014/P1/2]

A petrol gauge in a car indicates the volume V of the fuel in the tank. V is given by the angular deflection θ of the pointer on a dial.



Below are the calibration graphs for 4 different gauges. For low fuel levels in the tank, which gauge would be most sensitive?



21. [NYJC/Prelims 2014/P1/1]

Intensity of a wave I at a distance r can be determined using the equation

$$I = \frac{P}{4\pi r^2}$$

where P is the power of the source.

The fractional error in the measurement of power is a and that in the measurement of distance is b .

What is the fractional error in the calculated value of I ?

- A $a + 2b$ B $a - 8\pi b$ C $\frac{\Delta a}{a} - 2\frac{\Delta b}{b}$ D $\frac{\Delta a}{a} + 2\frac{\Delta b}{b}$

22. [PJC/Prelims 2014/P1/1]

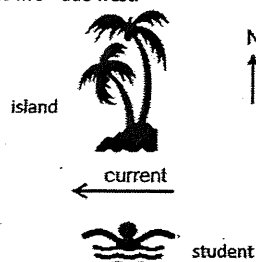
At temperatures close to 0 K, the specific heat capacity c of a particular solid is given by $c = bT^3$, where T is the thermodynamic temperature and b is a constant.

What are the units of constant b , expressed in SI base units?

- A $\text{m}^2 \text{s}^{-2} \text{K}^{-3}$ B $\text{m}^2 \text{s}^{-2} \text{K}^{-4}$ C $\text{kg m}^2 \text{s}^{-2} \text{K}^{-3}$ D $\text{kg m}^2 \text{s}^{-2} \text{K}^{-4}$

23. [RVHS/Prelims 2014/P1/1]

A student was shipwrecked a distance south of an island as shown below. If she can swim at a steady speed of 1.3 m s^{-1} in a current that flows at 1.0 m s^{-1} due west.



In what direction must she swim to make it to the island?

- A 38° east of north
B 38° west of north
C 40° east of north
D 50° east of north

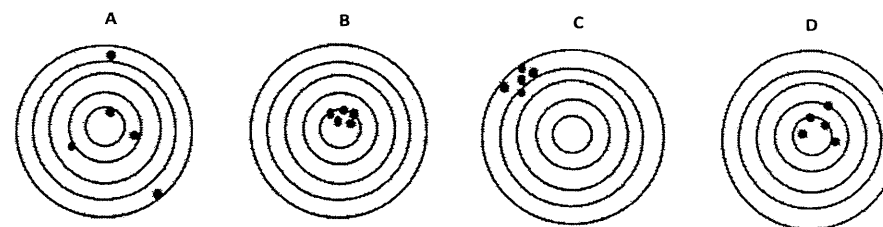
24. [SAJC/Prelims 2014/P1/1]

Which of the following correctly expresses the volt in terms of SI base units?

- A $\text{A } \Omega$ B W A^{-1} C $\text{kg m}^2 \text{s}^{-1} \text{A}^{-1}$ D $\text{kg m}^2 \text{s}^{-3} \text{A}^{-1}$

25. [SRJC/Prelims 2014/P1/1]

Shown below are four target boards from a shooting competition (the centre is awarded the highest points). Which one shows the most precise and least accurate shooting?



26. [SRJC/Prelims 2014/P1/2]

In the determination of the electro-chemical equivalent, z , of copper by electrolysis, the following equation is employed:

$$z = \frac{m_1 - m_2}{It}$$

$$m_1 \pm \Delta m_1 = (47.66 \pm 0.01) \times 10^{-3} \text{ kg}$$

$$m_2 \pm \Delta m_2 = (43.67 \pm 0.01) \times 10^{-3} \text{ kg}$$

$$I \pm \Delta I = (2.00 \pm 0.05) \text{ A}$$

$$t \pm \Delta t = (6000 \pm 1) \text{ s}$$

The value of z calculated from these readings will have a percentage uncertainty of

- A 2.5% B 2.8% C 3.0% D 3.3%

27. [TJC/Prelims 2014/P1/1]*

The unit of resistivity, expressed in terms of base units, is given by

$$\text{kg x}^3 \text{y}^{-2} \text{z}^{-3}$$

Which base units are x , y and z ?

	x	y	z
A	ampere	metre	second
B	metre	ampere	second
C	metre	second	ampere
D	second	ampere	metre

28. [TJC/Prelims 2014/P1/2]

Two lengths, a and b , are measured to be (51 ± 1) cm and (49 ± 1) cm respectively.
In which of the following quantities is the percentage uncertainty the largest?

- A $a + b$ B $a - b$ C a / b D ab

29. [TPJC/Prelims 2014/P1/1]

In an experiment, the external diameter d_1 and internal diameter d_2 of a glass tube are found to be (82 ± 2) mm and (53 ± 1) mm respectively. The percentage error in $(d_1 - d_2)$ is

- A 0.7% B 3% C 4% D 10%

30. [VJC/Prelims 2014/P1/1]

What is the approximate volume of a typical inflated rubber party balloon?

- A 0.001 m^3 B 0.01 m^3 C 0.1 m^3 D 1 m^3

31. [VJC/Prelims 2014/P1/2]

A force of (2.0 ± 0.1) N is applied to a mass of (3.00 ± 0.05) kg.
What is its acceleration?

- A $(0.67 \pm 0.02) \text{ m s}^{-2}$ B $(0.67 \pm 0.03) \text{ m s}^{-2}$ C $(0.67 \pm 0.04) \text{ m s}^{-2}$ D $(0.67 \pm 0.07) \text{ m s}^{-2}$

32. [YJC/Prelims 2014/P1/1]

Systematic and random errors have different properties.

P_1 : Has constant sign and magnitude

P_2 : Has varying sign and magnitude

Q_1 : Can possibly be eliminated

Q_2 : Cannot possibly be eliminated

R_1 : Will be reduced by taking average of repeated measurements

R_2 : Will not be reduced by taking average of repeated measurements

Which of the above properties are applicable to random errors?

- A P_1, Q_1, R_2 B P_1, Q_2, R_1 C P_2, Q_1, R_2 D P_2, Q_2, R_1

33. [YJC/Prelims 2014/P1/2]

Which of the following is the best estimate of the weight of a basketball?

- A 0.050 kg B 1000 mg C 5.0×10^{-6} MN D 1.0×10^{10} nN

34. [CJC/Prelims 2014/P1/3]

A car initially moving at 40 m s^{-1} due west makes a turn at a junction, and is now travelling at a speed of 30 m s^{-1} due north.

What is the car's change in velocity?

- A 10 m s^{-1} at a direction 53° east of north
B 10 m s^{-1} at a direction 37° west of north
C 50 m s^{-1} at a direction 53° east of north
D 50 m s^{-1} at a direction 37° west of north

II. Structured Questions**1. [NJC/Prelims 2014/P2/1]**

- a. The travelling microscope may be read to 0.1 mm .

Discuss whether it would be possible to use this apparatus to detect a variation of 1% in the diameter of the tube between two points, 25 mm apart. [2]

- b. To reduce random error, we can take the average of several readings of a quantity.

Explain why taking average of a few readings can reduce random error. [2]

- c. 'Velocity is a vector quantity, and kinetic energy of a body is equal to half its mass times the square of its velocity. Hence, kinetic energy must also be a vector.'

Comment on the correctness of this statement. [2]

[Total: 6]

2. [RVHS/Prelims 2014/P2/1]*

A student doing a project wired two identical resistors in parallel into a circuit. The manufacturer specifications for each resistor are: $390\ \Omega$; 5% tolerance (percentage uncertainty).

- a. Calculate the absolute uncertainty ΔR in the effective resistance of the two resistors in parallel. [3]

To verify the effective resistance is as calculated from manufacturer specifications, the student sets up the following circuit as shown in Fig. 2.1.

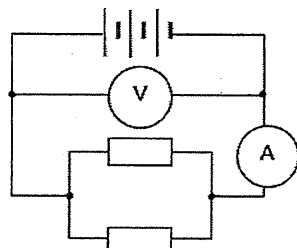


Fig. 2.1

The meter readings are as follows:

Voltmeter: $V = 6.02 \pm 0.01\text{ V}$

Ammeter: $I = 0.031 \pm 0.001\text{ A}$

- b. Calculate the absolute uncertainty ΔR_m of the effective resistance based on the meter readings. [3]
- c. Her project mate suggested another circuit instead for determining effective resistance.

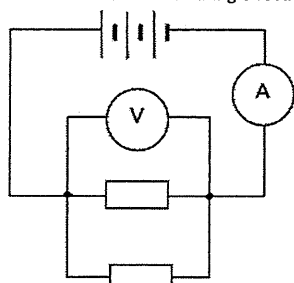


Fig. 2.2

Suggest under what condition(s) the circuit in Fig. 2.2 will be more appropriate than the circuit in Fig. 2.1. [1]

[Total: 7]

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
D	B	C	D	C	D	D	B	D	B	B	C	C	C	B

16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
B	C	C	B	C	A	B	D	D	C	C	B	B	D	B

31	32	33	34
C	D	C	C

- D (random uncertainty of ± 0.02) + (systematic uncertainty of ± 0.02)
- B estimate average speed as 10 m s^{-1} . apply $\frac{1}{2} mv^2$.
- C  Since particle is travelling anti-clockwise, the resultant (tangential) velocity shown above will occur in quadrant C.
- D Even though work done can be negative, it is a scalar since the negative does not represent its direction. Electric current flows along the wire and therefore has no fixed direction in space.
- C The absolute uncertainty of the diameter is $0.02 \times 5.0 = 0.1 \text{ cm}$.
The absolute uncertainty of the radius will be $0.1 \text{ cm} / 2 = 0.05 \text{ cm}$
The fractional uncertainty of the radius will be $0.05 \text{ cm} / 2.5 \text{ cm} = 0.02$.
- D Random error causes a scatter of points about an average which affects the precision.
Systematic error causes a shift in the value away from the true value which affects the accuracy.

- D Work = force $\times$ displacement, with units (N m)
e.m.f. $\mathcal{E}$, with unit (V)
current I with units (A)
- B Based on the equation, the graph is expected to have a non-zero y-intercept. However, the graph plotted shows a zero y-intercept. Therefore, the data is inaccurate. However, the scatter of the data is small, and most of the points lie close to the best-fit line. Therefore, the data is precise.
- D Rate of change of momentum = $\frac{dp}{dt} = F$ with units N.
Gradient of electric potential energy = electric force, with units N.
Impulse per unit time = $\frac{\text{impulse}}{\text{time taken}} = \frac{\text{change in momentum}}{\text{time taken}}$
= rate of change of momentum, with units N
Power over displacement: units N s^{-1}
- B $\frac{1}{\lambda} = R_H \left(1 - \frac{1}{n^2}\right) \rightarrow R_H = \frac{n^2}{\lambda(n^2 - 1)}$
 $R_H = \frac{2^2}{(122 \times 10^{-9})(2^2 - 1)} = 1.0929 \times 10^7 = 1.09 \times 10^7 \text{ m}^{-1}$ (1 d.p.)
uncertainty:
 $\frac{\Delta R_H}{R_H} = \frac{\Delta \lambda}{\lambda} = \frac{2}{122} \Rightarrow \Delta R_H = \frac{2}{122} (1.0929 \times 10^7) = 0.0179 \times 10^7 = 0.02 \times 10^7$ (1 sf)
 $R_H = (1.09 \pm 0.02) \times 10^7 \text{ m}^{-1}$
- B Estimating radius as 1 mm, $V = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi (1 \times 10^{-3})^3 = 4.2 \times 10^{-9} \text{ m}^3$
 $m = \rho V = 1000 \times 4.2 \times 10^{-9} = 4.2 \times 10^{-6} \text{ kg} = 4.2 \text{ mg}$
- C $F_x = \frac{\Delta X}{Y} = \frac{0.2}{15} = 0.0113$; $F_y = \frac{\Delta Y}{Y} = \frac{0.2}{30} = 0.00667$
 $Y - X = 15$ and $Y + X = 15$
 $\Delta F_{Y-X} = \Delta X + \Delta Y = \Delta F_{X+Y} = 0.2 + 0.2 = 0.4$
fractional uncertainty:
 $F_{Y-X} = \frac{\Delta F_{Y-X}}{Y - X} = \frac{0.4}{15} = 0.0267$ (largest)
 $F_{X+Y} = \frac{\Delta F_{X+Y}}{X + Y} = \frac{0.4}{45} = 0.00889$

13 C

$$R = \frac{pV}{nT} = \frac{(F/A)(V)}{nT}$$

Comparing units:

$$\text{units of } R = \frac{(\text{units of } F)(\text{units of } V)}{(\text{units of } A)(\text{units of } n)(\text{units of } T)} = \frac{(\text{kg m s}^{-2})(\text{m}^3)}{(\text{m}^2)(\text{mol})(\text{K})} = \text{kg m}^2 \text{ K}^{-1} \text{ mol}^{-1} \text{ s}^{-2}$$

14 C

$$P = I^2 R$$

$$\frac{\Delta P}{P} \times 100\% = 2 \frac{\Delta I}{I} \times 100\% + \frac{\Delta R}{R} \times 100\% = 2 \left(\frac{0.05}{3.50} \right) \times 100\% + \frac{2}{100} \times 100\% = 2.86 + 2 = 5\%$$

15 B

A: 300 K = 26.85 °C (room temperature)

$$\text{B: Diameter of soccer ball} = 22 \text{ cm, surface Area} = 4\pi \left(\frac{0.22}{2} \right)^2 = 0.15 \text{ m}^2$$

C: Typical laptop p.d. = 19 V and current = 4.22 A

D: Diameter of a pen = 1.0 cm; length of pen = 0.100 m

$$\text{Volume} = \pi \left(\frac{0.01}{2} \right)^2 (0.100) = 7.85 \times 10^{-6} \text{ m}^3$$

16 B

Using the parallelogram addition of vectors,

$$R = P + Q$$

$$S = P + (-Q)$$

$$T = (-P) + Q$$

17 C

$$P = I^2 R$$

$$\frac{\Delta P}{P} = 2 \frac{\Delta I}{I} + \frac{\Delta R}{R} = 2 \frac{0.05}{2.50} + 0.02 = 0.06 = 6\%$$

18 C

$$k = \frac{F}{r v}$$

$$\text{units of } k = \frac{\text{N}}{\text{m} \times \text{m s}^{-1}} = \frac{\text{kg m s}^{-2}}{\text{m} \times \text{m s}^{-1}} = \text{kg m}^{-1} \text{ s}^{-1}$$

19 B

Squaring both sides of the equation and comparing units:

$$\text{units of } v^2 = (\text{units of } a)(\text{units of } \lambda) = \frac{\text{units of } b}{(\text{units of } \rho)(\text{units of } \lambda)}$$

$$(\text{m s}^{-1})^2 = (\text{units of } a)(\text{m}) = \frac{\text{units of } b}{(\text{kg m}^{-3})(\text{m})}$$

$$\text{Hence, } \begin{aligned} \text{units of } a &= \text{m s}^{-2} \\ \text{units of } b &= \text{s}^{-2} \text{ kg} \end{aligned}$$

20 C

For high sensitivity to low fuel levels in the tank, the angle of deflection θ must be large for small values of V .

21 A

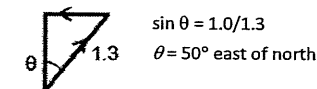
$$\frac{\Delta P}{P} = a, \quad \frac{\Delta r}{r} = b$$

$$\frac{\Delta I}{I} = \frac{\Delta P}{P} + 2 \frac{\Delta r}{r} = a + 2b$$

22 B

$$\text{Units of } b = \text{units of } c / K^3 = \text{m}^2 \text{ s}^{-2} \text{ K}^{-4}$$

23 D



24 D

$$V = \frac{W}{Q} = \frac{W}{It} = \frac{mas}{It}$$

$$\text{units of } V = \frac{\text{kg m s}^{-2} \text{ m}}{\text{A s}} = \text{kg m}^2 \text{ s}^{-3} \text{ A}^{-1}$$

25 C

Most precise and least accurate means that shots are close to each other and the average location of the shots are the furthest from the centre of the target board.

26 C

$$\text{Let } A = m_1 - m_2$$

$$\text{Let } B = It$$

$$\frac{\Delta z}{z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

$$\frac{\Delta A}{A} = \frac{\Delta m_1 + \Delta m_2}{m_1 - m_2} \quad \frac{\Delta B}{B} = \frac{\Delta I}{I} + \frac{\Delta t}{t}$$

$$\frac{\Delta z}{z} = \frac{\Delta m_1 + \Delta m_2}{m_1 - m_2} + \frac{\Delta I}{I} + \frac{\Delta t}{t} = \frac{0.02}{47.66 - 43.47} + \frac{0.05}{2} + \frac{1}{6000} = 0.03$$

$$\frac{\Delta z}{z} \times 100\% = 0.03 \times 100\% = 3.0\%$$

27 B

Resistivity ρ is measured in Ohm Meters or $\text{kg m}^3 \text{ A}^{-2} \text{ s}^{-3}$.

28 B

percentage uncertainty the largest $\frac{\Delta x}{X}$, X should be the smallest. Therefore $a - b$.because $a - b = 2 \text{ cm}$ and $2/2 \times 100 = 100\%$!

29 D $d_1 - d_2 = 82 - 53 = 29 \text{ mm}$
 $\Delta(d_1 - d_2) = \Delta d_1 + \Delta d_2 = 2 + 1 = 3$
 percentarge error = $\frac{\Delta(d_1 - d_2)}{d_1 - d_2} \times 100\% = \frac{3}{29} \times 100\% = 10.3\%$

- 30 B Option A gives a radius of 6 cm
 Option B gives a radius of 13 cm
 Option C gives a radius of 29 cm
 Option D gives a radius of 62 cm
 So B is the most sensible answer.

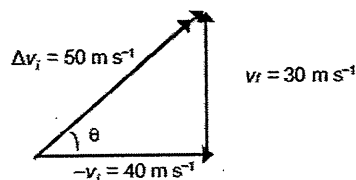
31 C $F = ma \Rightarrow a = F/m$
 $\Delta a = a \left(\frac{\Delta F}{F} + \frac{\Delta m}{m} \right) = 0.67 \left(\frac{0.1}{2.0} + \frac{0.05}{3.00} \right) = 0.04 \text{ m s}^{-2}$

- 32 D Refer to notes for definitions.

- 33 C Weight = gravitational force acting on the ball (in Newton)
 $W = mg$
 Option A and B: units used are for mass
 Option C: 5 N $\rightarrow m \sim 500\text{g}$
 Option D: 10 N $\rightarrow m \sim 1\text{kg}$ (too heavy!)

34 C $\Delta v = v_f - v_i = v_f + (-v_i)$
 $\Delta v = \sqrt{40^2 + 30^2} = 50 \text{ m s}^{-1}$
 $\theta = \tan^{-1} \left(\frac{30}{40} \right) = 36.9^\circ$

Thus, the direction of the car's change in velocity is 37° north of east or 53° east of north.



Structured Questions

1. [YJC/Prelims 2014/P1/2]

- a. $0.01 \times 25 = 0.25 \text{ mm} = 0.3 \text{ mm}$
 Hence, not possible as the instrument cannot read up to the 2nd dp.
- b. Taking average of a few readings could reduce the scatter of the readings from one another. Hence, random error is reduced since random error is a measure of the scatter of the data points.
- c. The statement is incorrect. The kinetic energy of a body is equal to half its mass times the square of the value of its velocity. Since the calculation consists of the multiplication of scalar values, the result is a scalar value and not a vector.

2. [RVHS/Prelims 2014/P2/1]*

- a. actual uncertainty for each resistor = 5% of $R = 19.5 = 20 \Omega$ (to 1 s.f.)
 range of possible values for each resistor = $(390 \pm 20) \Omega = 370$ to 410Ω

$$\text{Maximum possible resistance} = \left[\frac{1}{410} + \frac{1}{410} \right]^{-1} = 205 \Omega$$

$$\text{Minimum possible resistance} = \left[\frac{1}{370} + \frac{1}{370} \right]^{-1} = 185 \Omega$$

$$\text{Total } \Delta R = \frac{\text{max} - \text{min}}{2} = \frac{205 - 185}{2} = 10 \Omega$$

Alternative method:

actual uncertainty for each resistor = 5% of $R = 19.5 = 20 \Omega$ (to 1 s.f.)

effective resistance = $\frac{1}{2}$ resistor

$\Delta R = \frac{1}{2}$ actual uncertainty for resistor = $\frac{1}{2} (20) = 10 \Omega$ (to 1 s.f.)

- b. $R = V/I = 194.1935 \Omega$

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$

$$\frac{\Delta R}{194.19} = \frac{0.01}{6.02} + \frac{0.001}{0.031}$$

$$\Delta R = 7 \Omega$$

- c. When the resistance of the ammeter is large compared to the effective resistance. (If the circuit in Fig. 2.1 is used, the voltmeter will not give correct potential difference of the resistances.)

